Practice Problems on K-Nearest Neighbors (KNN)

Problem Set

Problem 1: 1-NN Decision Boundary (2D) You are given the following labeled points:

Negative: $(1, 2), (3, 4), (5, 1)$ Positive: $(2, 5), (4, 3), (6, 2)$

(a) Draw the 1-NN decision boundary (Voronoi diagram restricted to positive vs. negative).

(b) Classify the test point $(3, 2)$.

Problem 2: k-NN with $k = 3$ Training data points:

Negative: $(1, 1), (2, 3), (4, 1)$ Positive: $(3, 4), (5, 2), (6, 3)$

(a) Plot the training points.

(b) Using $k = 3$, classify the unknown test point $(3, 2)$. Show distances and majority voting.

Problem 3: Changing k Effect Given the following dataset:

Negative: $(1, 1), (2, 2), (2, 4)$ Positive: $(4, 1), (5, 3), (6, 2)$

(a) For $k = 1$, classify the test point $(3, 3)$.

(b) For $k = 5$, classify the same test point.

(c) Discuss how increasing k affects the bias–variance tradeoff.

Problem 4: Weighted k-NN You are given training data:

Negative: $(2, 1), (3, 2)$ Positive: $(4, 2), (5, 3)$

(a) For test point $(3.5, 2.5)$, compute distances to all points.

(b) Classify using **weighted 3-NN**, where weight $= 1/d$ (inverse of distance).

(c) Compare with unweighted 3-NN result.

Problem 5: Real-world Inspired Suppose we classify fruits based on features: - Negative = "Apple", Positive = "Orange" - Features: (Sweetness, Weight)

Data:

$$\text{Apple: } (2, 150), (3, 140), (4, 160) \qquad \text{Orange: } (6, 180), (7, 200), (8, 190)$$

(a) Normalize the features (z-score normalization).

(b) Classify the unknown fruit $(5, 170)$ using 1-NN.

(c) Repeat classification using 3-NN.

Practice Problems on Confusion Matrix and Classification Metrics

Problem 1: Email Spam Classifier

A spam classifier was tested on 1000 emails. The results are:

- Correctly classified 400 spam emails as spam
- Incorrectly classified 50 spam emails as non-spam
- Correctly classified 480 non-spam emails as non-spam
- Incorrectly classified 70 non-spam emails as spam

(a) Construct the confusion matrix.

(b) Compute accuracy, precision, recall, and F1-score.

Problem 2: Medical Test

A diagnostic test for a disease was applied to 500 patients, of which 120 had the disease. The test results were:

$$\text{TP} = 90, \quad \text{FN} = 30, \quad \text{FP} = 60, \quad \text{TN} = ?$$

(a) Complete the confusion matrix.

(b) Compute sensitivity (recall), specificity, and accuracy.

(c) Discuss whether the test is better at detecting sick patients or healthy patients.

Problem 3: Fake News Detection

A fake news detector was evaluated on 800 news articles, with the following outcomes:

$$\text{TP} = 320, \quad \text{FN} = 80, \quad \text{FP} = 100, \quad \text{TN} = 300$$

(a) Draw the confusion matrix.

(b) Compute precision, recall, specificity, and F1-score.

(c) Interpret the trade-off between catching fake news and mislabeling real news.

Problem 4: Credit Card Fraud Detection

A bank tested a fraud detection model on 10,000 transactions. Only 100 were fraudulent. Results:

$$\text{TP} = 85, \quad \text{FN} = 15, \quad \text{FP} = 200, \quad \text{TN} = 9700$$

(a) Construct the confusion matrix.

(b) Compute accuracy, precision, recall, and F1-score.

(c) Why is accuracy misleading in this case? Which metric is more appropriate?

Problem 5: Sentiment Classifier

A sentiment analysis model predicts whether a movie review is positive or negative. On a test set of 500 reviews, it produced the following confusion matrix:

	Predicted Positive	Predicted Negative
Actual Positive	180	70
Actual Negative	50	200

(a) Verify the totals and compute accuracy.

(b) Compute precision, recall, and F1-score for the positive class.

(c) If the model is deployed for recommending movies, why might recall be more important than precision?